01 - Gap Analysis: marcosandbox vs FinOps Hubs Reference

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Phase: Assessment
Audience: [architects, finops-analysts, engineers]
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Reference: Microsoft FinOps Toolkit

Overview

This document compares the current marcosandbox FinOps infrastructure against the Microsoft FinOps Hubs reference architecture to identify gaps and prioritize remediation efforts. The FinOps Hubs pattern provides enterprise-grade cost management with centralized data ingestion, normalization, allocation, and reporting.

Gap Summary: 7 critical components identified, 3 present, 4 missing or partial.

FinOps Hubs Reference Architecture (Minimum Components)

Tier 1: Data Collection & Landing

- **ADLS Gen2 Storage** with hierarchical namespace
  - Container: `raw/` (landing zone for exports)
  - Container: `processed/` (normalized/enriched data)
  - Container: `archive/` (long-term retention)
- **Event Grid** integration (Storage → trigger ingestion)
- **Cost Management Exports** (daily/monthly, ActualCost + Amortized)

Tier 2: Ingestion & Transformation

- **Azure Data Factory** (or Synapse/Fabric) pipelines
  - Copy activity: Export CSV → Raw storage
  - Data flow: Normalize, enrich, tag parsing
- **Azure Data Explorer (ADX)** cluster + database
  - Table: `raw_costs` (direct CSV ingestion)
  - Table: `normalized_costs` (cleansed, typed columns)
  - Table: `apim_usage` (for attribution)
- **Ingestion Mappings** (CSV → ADX schema)

Tier 3: Attribution & Allocation

- **APIM Integration** (telemetry with custom headers)
  - Policy: Inject `x-costcenter`, `x-caller-app`, `x-environment`
  - Diagnostics: Forward to Log Analytics/App Insights

- **ADX Allocation Views** (KQL functions)
  - Join cost data with usage telemetry by timestamp/resource
  - Allocate spend to consuming applications/cost centers

Tier 4: Reporting & Governance

- **Power BI Workspace** with KQL direct query
  - Template: Cost trends, resource group breakdown, tag compliance
- **Azure Policy** enforcement
  - Require tags: `CostCenter`, `Application`, `Environment`
  - Enforce export configuration at subscription level
- **RBAC/Private Endpoints** (enterprise security)
  - Managed identities with least-privilege roles
  - Private endpoints for storage, ADX, ADF

Current State Mapping (Evidence-Based)

☑ Components Present

Component	Status	Evidence	Notes
ADLS Gen2 Storage	☑ Present	MARCO-INVENTORY L107	marcosandboxfinopshub (StorageV2, dfs endpoint enabled)
Event Grid systemTopic	☑ Present	MARCO-INVENTORY L157	marcosandboxfinopshub-52dd1c15... configured
Cost Exports (Active)	☑ Active	14-az-finops README L35-36	Daily exports for both subscriptions (Feb 16, 2026)
Data Factory	⚠ Partial	MARCO-INVENTORY L145	marco-sandbox-finops-adf exists, pipelines UNKNOWN
APIM	⚠ Partial	MARCO-INVENTORY L136	marco-sandbox-apim exists, policies UNKNOWN
App Insights	☑ Present	MARCO-INVENTORY L137	marco-sandbox-appinsights configured

❌ Components Missing

Component	Status	Impact	Priority
ADX Cluster + Database	❌ Missing	Cannot perform analytics, KQL queries, or Power BI integration	CRITICAL
Storage Container Structure	⚠ Incomplete	Only `costs/` present, need `raw/`, `processed/`, `archive/`	HIGH
ADF Ingestion Pipelines	❌ Unknown/Missing	No automated CSV → ADX ingestion	CRITICAL

Component	Status	Impact	Priority
APIM Attribution Policies	❌ Unknown	No cost allocation by caller/app	CRITICAL
ADX Ingestion Mappings	❌ Missing	Cannot ingest cost CSV into ADX tables	CRITICAL
Power BI Integration	❌ Missing	No reporting capability	HIGH
Azure Policy (Tags)	❌ Missing	No tag enforcement	MEDIUM
Private Endpoints	⚠ Unknown	Network security posture unclear	MEDIUM

Detailed Gap Analysis

Gap 1: Azure Data Explorer (ADX) Cluster [CRITICAL]

Current State: No ADX cluster found in inventory snapshot.

Evidence: MARCO-INVENTORY lists 24 marco-sandbox resources; no ADX/Kusto resources present.

Impact:

- Cannot perform large-scale cost analytics (CSV exports not queryable at scale)
- Power BI cannot use KQL direct query (forced to import mode with refresh limits)
- No near-real-time allocation views (joins must be done in Power BI desktop)

Recommended Action:

- Deploy ADX cluster in canadacentral (Dev SKU: 2 nodes, Engine D11_v2 ~$300/month)
- Create database `finopsdb` with tables: `raw_costs`, `normalized_costs`, `apim_usage`
- Configure ingestion endpoint and managed identity

Acceptance Criteria:

- ADX cluster provisioned and query endpoint accessible
- Database and schema created via IaC (Bicep/Terraform)
- Sample CSV ingested successfully and KQL query returns results

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Gap 2: Storage Container Hierarchy [HIGH]

Current State: Single container `costs/` with subscription subdirectories.

Evidence: Export paths documented in 14-az-finops README: `marcosandboxfinopshub, container=costs, directory={SubscriptionName}`

Impact:

- No separation between raw landing zone and processed data
- Cannot implement lifecycle policies per processing stage
- Difficult to track data lineage (raw → processed → aggregated)

Recommended Structure:

```
marcosandboxfinopshub/
├── raw/costs/{subscription}/{YYYY}/{MM}/         # Landing zone for exports
├── processed/costs/{subscription}/{YYYY}/{MM}/   # Normalized, enriched CSVs
├── archive/costs/{subscription}/{YYYY}/          # Archived after ADX ingestion
└── checkpoint/                                   # ADF pipeline state/manifest
```

Acceptance Criteria:

- Containers created and accessible via `az storage container list`
- Lifecycle policy configured: move to Cool after 90 days, Archive after 180 days
- ADF pipeline writes manifest to `checkpoint/` after successful ingestion

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Gap 3: ADF Ingestion Pipelines [CRITICAL]

Current State: ADF factory `marco-sandbox-finops-adf` exists but pipeline configuration unknown.

Evidence: MARCO-INVENTORY L145

Impact:

- No automated ingestion from Blob → ADX
- Manual effort required to download and process exports
- Event Grid triggers have no consumer (wasted events)

Required Pipelines:

Pipeline Name	Trigger	Activities	Output
`ingest-costs-to-adx`	Event Grid (blob created)	1. Decompress CSV 2. Validate schema 3. ADX copy activity 4. Move to processed/	ADX table populated
`backfill-historical`	Manual/Scheduled	1. List blobs in raw/ 2. Iterate and ingest 3. Track progress in checkpoint/	Historical data in ADX
`enrich-and-normalize`	Scheduled (daily 2 AM)	1. Read processed/ 2. Parse tags 3. Apply enrichment rules 4. Write to normalized_costs	Enriched dataset

Acceptance Criteria:

- Pipelines deployed via IaC (ARM template in repo)
- Test execution with sample CSV successful

- Monitoring alerts configured for pipeline failures

Gap 4: APIM Cost Attribution [CRITICAL]

Current State: APIM resource `marco-sandbox-apim` exists with 2 deployed APIs (echo-api, eva-jp-query-api at path `/eva-jp/v1`), but comprehensive policy configuration for cost attribution headers is UNKNOWN. An **EVA-JP API Inventory Project** is underway to catalog and attribute **50+ APIs** (38+ active endpoints confirmed) from `I:\EVA-JP-reference-0113\app\backend\`.

Evidence:

- MARCO-INVENTORY L136
- Inventory baseline (2026-02-17 09:09:32): 2 APIs deployed (echo-api, eva-jp-query-api)
- EVA-JP Backend: 38+ active endpoints (33 @app.get/post/put/delete + 5 @router endpoints in sessions.py)
- Source: `I:\EVA-JP-reference-0113\app\backend\app.py` (FastAPI application, 2144 lines)

Impact:

- No ability to allocate Azure spend to consuming applications
- Cannot implement showback/chargeback to internal teams (50+ APIs require cost visibility)
- Usage telemetry not correlated with cost data
- **50+ EVA-JP APIs** lack granular cost attribution across environments (dev/test/production)

Required Implementation:

APIM Policy Snippet (inbound):

```
<policies>
    <inbound>
        <!-- Extract cost center from JWT claim or header -->
        <set-variable name="costCenter"
value="@(context.Request.Headers.GetValueOrDefault("x-costcenter", "UNKNOWN"))" />
        <set-variable name="callerApp"
value="@(context.Request.Headers.GetValueOrDefault("x-caller-app",
context.Request.Headers.GetValueOrDefault("x-client-id", "UNKNOWN")))" />
        <set-variable name="environment"
value="@(context.Request.Headers.GetValueOrDefault("x-environment",
"production"))" />

        <!-- Add to App Insights custom dimensions -->
        <set-header name="x-eva-costcenter" exists-action="override">
            <value>@((string)context.Variables["costCenter"])</value>
        </set-header>
        <set-header name="x-eva-caller-app" exists-action="override">
            <value>@((string)context.Variables["callerApp"])</value>
        </set-header>
    </inbound>
    <backend>
        <base />
    </backend>
```

```
    <outbound>
        <base />
    </outbound>
</policies>
```

Diagnostic Configuration:

- Enable diagnostics for **all APIs** (50+ EVA-JP APIs from inventory project) → App Insights `marco-sandbox-appinsights`
- Include custom dimensions (headers) in telemetry
- Configure sampling (100% for cost attribution accuracy)
- **EVA-JP API Catalog Integration**: Map 38+ active endpoints to cost center tags

Acceptance Criteria:

- Policy applied to all APIs (or base policy covering 50+ APIs)
- Test request shows custom dimensions in App Insights logs
- ADX ingestion pipeline extracts headers from telemetry
- **80%+ of EVA-JP API requests** have cost center attribution (per KPI in 04-backlog.md)

Gap 5: ADX Ingestion Mappings [CRITICAL]

Current State: No ADX cluster means no ingestion mappings.

Required Schema (Cost CSV → ADX):

```
.create table raw_costs (
    BillingAccountId: string,
    BillingAccountName: string,
    BillingPeriodStartDate: datetime,
    BillingPeriodEndDate: datetime,
    BillingProfileId: string,
    BillingProfileName: string,
    AccountOwnerId: string,
    AccountOwnerName: string,
    SubscriptionId: string,
    SubscriptionName: string,
    ResourceGroup: string,
    ResourceLocation: string,
    Date: datetime,
    ProductName: string,
    MeterCategory: string,
    MeterSubCategory: string,
    MeterId: string,
    MeterName: string,
    MeterRegion: string,
    UnitOfMeasure: string,
    Quantity: real,
    EffectivePrice: real,
    CostInBillingCurrency: real,
```

```
    CostCenter: string,
    ConsumedService: string,
    ResourceId: string,
    Tags: dynamic,
    OfferId: string,
    IsAzureCreditEligible: string,
    PartNumber: string,
    PayGPrice: real,
    PricingModel: string,
    ResourceName: string,
    ServiceFamily: string,
    UnitPrice: real,
    // Additional columns as needed
    IngestionTime: datetime
)

.create table raw_costs ingestion csv mapping 'CostExportMapping'
'['
'{"column":"Date","DataType":"datetime","Properties":{"Ordinal":"0"}},'
'{"column":"SubscriptionName","DataType":"string","Properties":{"Ordinal":"1"}},'
'{"column":"ResourceGroup","DataType":"string","Properties":{"Ordinal":"2"}},'
// ... (continue for all 55+ columns)
']'
```

Acceptance Criteria:

- Schema created in ADX database
- CSV ingestion mapping tested with sample export
- Data types validated (no conversion errors)

Gap 6: Power BI Integration [HIGH]

Current State: No Power BI workspace or reports configured.

Required Components:

- Power BI workspace (shared capacity or Premium)
- ADX connection (KQL direct query, not import mode)
- Report templates:
    1. **Cost Trend Dashboard** (daily/weekly/monthly aggregates)
    2. **Resource Group Breakdown** (top 10 by spend)
    3. **Tag Compliance Report** (untagged resources, missing cost center)
    4. **APIM Allocation Report** (spend by caller app/cost center)

Acceptance Criteria:

- Workspace created and access granted to FinOps team
- Sample PBIX connected to ADX and renders data
- Scheduled refresh configured (if using import mode for aggregates)

Gap 7: Governance (Policies & RBAC) [MEDIUM]

Current State: Tags present on marco-sandbox resources but enforcement unknown.

Required Azure Policies:

Policy	Effect	Scope	Parameters
Require CostCenter tag	Deny	Subscription	List of resource types
Require Environment tag	Audit	Subscription	Valid values: dev,stage,prod
Enforce Cost Export	DeployIfNotExists	Subscription	Export to marcosandboxfinopshub
Restrict Storage Firewall	Deny	Sandbox RG	Allow only VNet/Private Endpoints

RBAC Requirements (Managed Identities):

Identity	Scope	Role	Purpose
`mi-finops-adf`	marcosandboxfinopshub	Storage Blob Data Contributor	Read exports, write processed
`mi-finops-adf`	marco-finops-adx	Database Ingestor	Ingest to ADX tables
`mi-finops-powerbi`	marco-finops-adx	Database Viewer	Power BI query access

Acceptance Criteria:

- Policies assigned and compliance dashboard shows results
- Managed identities created and role assignments validated
- Test: Create resource without CostCenter tag → denied

Priority Matrix

Gap	Impact	Effort	Priority	Start Phase
ADX Cluster	CRITICAL	HIGH	P0	Phase 2
ADF Pipelines	CRITICAL	HIGH	P0	Phase 2
APIM Attribution	CRITICAL	MEDIUM	P0	Phase 3
Ingestion Mappings	CRITICAL	MEDIUM	P0	Phase 2
Storage Hierarchy	HIGH	LOW	P1	Phase 1
Power BI	HIGH	MEDIUM	P1	Phase 3
Governance	MEDIUM	MEDIUM	P2	Phase 4

Recommended Remediation Sequence

Phase 1: Foundation (Weeks 1-2)

- Create storage container structure (`raw/`, `processed/`, `archive/`)
- Configure Event Grid subscription to ADF (if not present)
- Validate current exports landing correctly

Phase 2: Analytics Infrastructure (Weeks 3-4)

- Deploy ADX cluster + database
- Create ingestion mappings and schema
- Deploy ADF pipelines for ingestion
- Test end-to-end: Export → Blob → ADF → ADX

Phase 3: Attribution & Reports (Weeks 5-7)

- Implement APIM attribution policies and diagnostics
- Create ADX telemetry ingestion pipeline
- Backfill historical cost data (12 months)
- Deploy Power BI reports

Phase 4: Hardening (Weeks 8-9)

- Deploy Azure Policies for tag enforcement
- Enable private endpoints for storage and ADX
- Implement CI/CD for IaC modules
- Conduct security review

Cost Estimate (Incremental Monthly)

Resource	SKU/Config	Monthly Cost (CAD)
ADX Cluster	Dev (2 nodes D11_v2)	~$300
ADF	~10 pipeline runs/day	~$20
Storage (additional)	Processed/Archive containers	~$10
Power BI	Shared workspace (existing)	$0
Total Incremental		**~$330/month**

ROI: With $297K/month combined spend, even 0.5% optimization (~$1,485/month) pays for infrastructure 4.5x over.

Change Log

Date	Author	Change
2026-02-17 08:20 AM ET	Marco Presta	Initial gap analysis with evidence mapping and remediation plan

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Document Status: Draft (pending inventory script execution for UNKNOWN validation)
Next Review: 2026-02-24 (after Phase 1 completion)